

TrES-2b

R-1055

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_151022 · created 2026-07-30T10:52:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457317.67037 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1221 +/- 0.0096
Transit depth (Rp/Rs)^2	1.49 +/- 0.24 [%]
Orbital Inclination (inc)	87.25 +/- 0.69
Airmass coefficient 1 (a1)	0.992 +/- 0.012
Airmass coefficient 2 (a2)	0.0125 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [208, 431]
Optimal Aperture	3.63
Optimal Annulus	9.96
Transit Duration (day)	0.1039 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1221 $\pm$ 0.0096 vs 0.1254 (deviation 0.16 σ)
Duration fit vs published	0.1039 d vs 0.0746 d (deviation 4.95 σ)
Mid-time O-C	1.0 min
Depth SNR	6.0
Residual scatter	1.24 %

Light curve

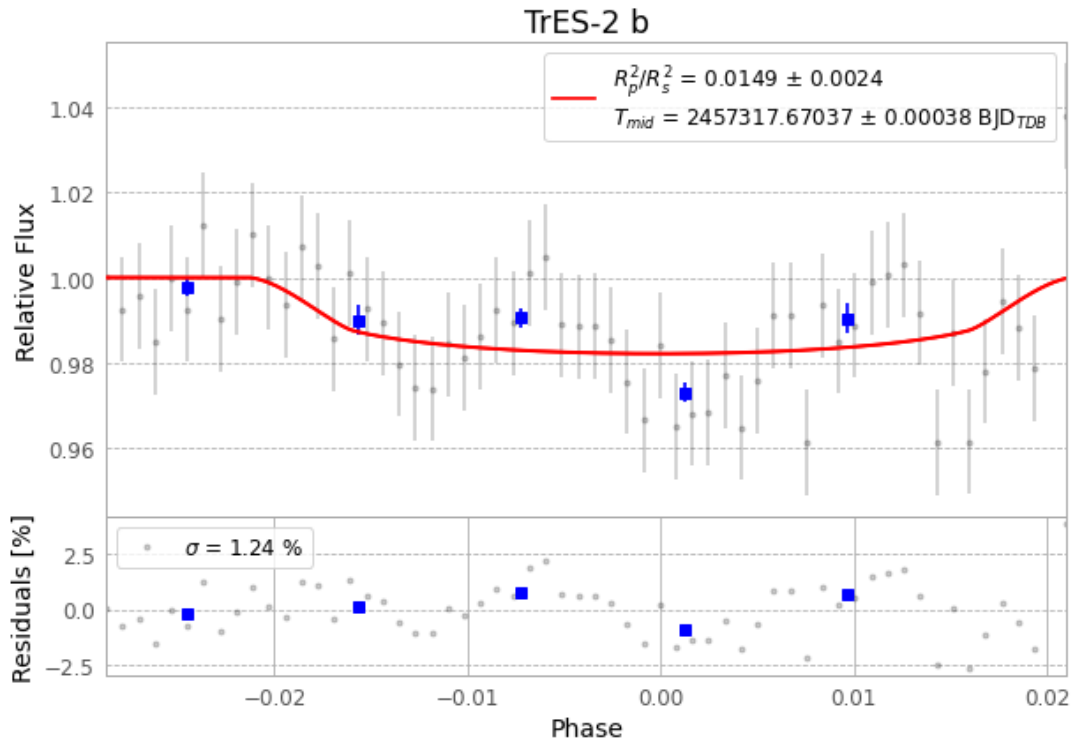


Figure 1 — FinalLightCurve_TrES-2 b_2015-10-21.png (SHA-256 a3aa7fe3f5c9a6c2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 298], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 55ce34b90f680edc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2151022022112.FITS → TRES-2151022055112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-151022.log (7565eba06b58...), FinalLightCurve_TrES-2 b_2015-10-21.png (a3aa7fe3f5c9...), FinalLightCurve_TrES-2 b_2015-10-21.csv (4f8408c2ba32...)

Compute resources

Wall time 310.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 10:52Z.